Integrated Approaches to Reliable and Robust CNS in Advanced Air Mobility

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1 State of the Art

1.1 Modelling the communication network

ICAO suggests a multi-link communication system for meeting future mobility requirements, of which AAM is a integral part (cite). As important it is to use a multi-link architecture, there needs to be intelligent systems in place to constantly monitor and manage its operations, especially in a complex use-case such as AAM. A successful implementation of AAM concepts would lead to an ever increasing density of fast moving users especially in the vertical direction. Take-off, landing and flying in the VLL necessitate a highly robust signal link with minimum interference. A multi-link system involves multiple sub-systems working together, utilizing different frequency bands, modulation techniques and update rates. This heterogeneity is complex to deal with in terms of handover between links and varying latency, leading to desynchronized data transmission. Moreover, it is crucial to consider the increased power requirements from the large number of additional node in the network.

An intelligent communication system is therefore required, that is capable of consistent monitoring of critical network performance indicators, and autonomously making decisions to ensure optimum operational efficiency for a high degree of robustness and reliability. Such a network can be imagined as having two operational levels. The first being the logical level deals with network management. Critical operations being the selection of optimal link, signal encryption, protocol selection,..... The second physical level as the name suggests, consists of the physical elements of the network. Namely transmitters, receivers, relays, other devices and the propagation environment itself. Towards mitigating the challenges to a multi-link architecture for AAM, it is proposed to include metamaterial-based devices such as the reconfigurable intelligent surface (RIS) as network nodes.

Metamaterials are engineered materials, custom-tailored for specific purposes. By definition they exercise control over whatever use-case they are designed for. Despite being critical for optimal resource utilization, there is a severe lack of control over EM wave propagation in the communication network, especially for a highly dynamic use-case such as AAM. Literature showcases several promising uses of metamaterials and their applications in designing a robust multi-link communication system towards reduced latency and desync, improved security, scalability, and optimum resource (energy and material) utilization (add separate refs for each, or a few for all of them ?). Metamaterial-based devices may directly act as one, or be included in any of the components within the physical level to enhance performance.

For safe, de-centralized and autonomous operation of an intelligent communication system it is critical that the two levels exchange information regularly and modify operational parameters accordingly. Numerical models like the FEM translate available information from the physical level such as geometry, material properties of devices and the propagation environment, topological placement of communication nodes, frequency and polarization of the utilized EM wave to time-dependent channel state information that could be used in the logical level to make smart decisions.